

StressFold: Perturbation-response profiling for tabular model generalization

Roman Mirosenski

github.com/RomanMski

Abstract

Overfitting is usually reduced to a train-test gap. Robustness is reported as a second accuracy number, and synthetic data are counted as extra evidence. These quantities answer different questions, and StressFold is a model-agnostic reference protocol for tabular classification and regression that keeps them apart. It estimates clean generalization under an explicit split policy, traces paired perturbation-response curves under declared measurement or distribution shifts, and defines falsification tests that rerun the complete workflow. The shipped implementations go less far and expose descriptive null comparisons instead. Synthetic replicas are admitted only as audited stress instruments. They do not enlarge the effective sample size, and they do not substitute for real holdout observations. The reference protocol specifies estimands, leakage boundaries, perturbation operators, paired Monte Carlo aggregation, and generator diagnostics. Controlled experiments then expose three failure modes that a single score conceals. An interpolating tree with poor population loss can improve under randomized measurement noise. Common perturbation draws materially reduce comparison variance. Replicas with accurate one-dimensional marginals can destroy class-conditional dependence and downstream utility. The browser and Python code cover the core split and perturbation workflow deterministically and locally. Full permutation inference and generator auditing are documented extensions that the current software does not claim to implement.

Keywords: generalization, robustness, permutation tests, synthetic data, tabular learning, Monte Carlo.

1 Technical summary

StressFold is built around a negative claim: no finite collection of resamples or generated rows can certify that a fitted model will generalize to every future environment. What can be measured is narrower than that, and still worth having. Given a complete learning algorithm, a split policy, a loss, and a family of perturbations, the protocol produces an evidence profile with three non-interchangeable components:

1. **Generalization** estimates predictive risk on observations excluded from all fitting and selection.

2. **Robustness** estimates the change in risk under a named perturbation mechanism and severity scale.
3. **Falsification** asks whether the entire observed workflow could produce its result under a specified null.

Contribution, and what is not claimed. The contribution is a synthesis and reproducible implementation of established validation ideas, together with controlled counterexamples that show why their outputs should not be collapsed into a single score. There is no new statistical theorem here. Nothing below amounts to a universal test for overfitting or a guarantee of performance once the deployment distribution moves.

Release 0.2 implements repeated holdout, four stress operators, and descriptive null diagnostics. Full-workflow permutation inference, group- and time-aware splitting, and generator auditing remain reference-protocol components rather than current software claims.

The distinctions are operational. Cross-validation estimates the performance of an algorithm trained on comparable unseen samples, which need not equal the conditional risk of the single final fit. Fold-level standard errors can also come out much too small (Bates et al., 2024). Common-corruption benchmarks show why a response curve carries more information than a single shifted test set (Hendrycks and Dietterich, 2019), and natural-domain benchmarks show that convenient synthetic shifts do not replace real changes across time, sites, devices, or populations (Koh et al., 2021).

The output is therefore a set of losses, paired differences, intervals, and explicit assumptions, rather than a composite traffic light. A model may generalize well on clean data and still be sensitive to measurement error. A highly adaptive model may fit random labels and still exploit genuine structure (Zhang et al., 2017). A generator may match marginals while omitting the dependence that makes prediction possible. Combining these outcomes would erase the information the audit exists to expose.

2 Clean fit and perturbation response can disagree

Figure 1 gives the central controlled example. The data-generating process (DGP) is known, so population Brier

loss and label-preserving measurement error can be evaluated without assuming that arbitrary noisy rows retain their labels. Increasing tree depth drives training loss to zero but raises population loss after depth two. Under perturbation, the depth-three model steadily loses skill. The interpolating tree instead improves: random input displacement moves some extreme, sample-specific leaf probabilities toward a useful randomized decision. The improvement is real for this perturbation distribution, but it does not rescue the tree’s poor clean population loss.

This example also fixes the interpretation of a response curve. It is conditional on an operator, its units, its support constraints, and a label semantics. A curve can diagnose brittleness or accidental smoothing. It cannot identify the deployment distribution unless the operator is itself a credible model of that distribution.

These simulations are constructive counterexamples under stated data-generating processes, not comparative benchmark evidence of general superiority.

3 Audit object and estimands

3.1 The complete algorithm is the unit under test

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be sampled from population P . The audited algorithm A maps a training sample to a predictor,

$$f_{\mathcal{T}} = A(\mathcal{T}).$$

Here A includes imputation, encoding, scaling, feature selection, hyperparameter search, probability calibration, threshold selection, and fitting. Treating only the final estimator as A leaves selection leakage outside the audit.

An outer split policy produces $(\mathcal{T}_b, \mathcal{H}_b)$, $b = 1, \dots, B$. The policy is part of the scientific specification. Stratified random splitting is appropriate only for exchangeable rows. Repeated entities require grouped splits, temporal prediction requires rolling-origin splits, and dependent sequences require blocks. Preprocessing statistics, perturbation calibration, and synthetic generators are fit using $\mathcal{T}_b$ only.

3.2 Generalization

For a fitted predictor $f_{\mathcal{T}}$, population risk under loss ℓ is

$$\mathcal{R}_P(f_{\mathcal{T}}) = \mathbb{E}_{(X,Y) \sim P}[\ell(f_{\mathcal{T}}(X), Y) \mid \mathcal{T}]. \quad (1)$$

The clean audit estimate is $\hat{\mathcal{R}}_{b,0} = |\mathcal{H}_b|^{-1} \sum_{i \in \mathcal{H}_b} \ell(f_{\mathcal{T}_b}(x_i), y_i)$. The descriptive generalization gap

$$G_b = \hat{\mathcal{R}}_{b,0} - |\mathcal{T}_b|^{-1} \sum_{i \in \mathcal{T}_b} \ell(f_{\mathcal{T}_b}(x_i), y_i) \quad (2)$$

is useful but not sufficient. A small gap can accompany uniformly poor predictions, and a flexible model can generalize despite a near-zero training loss. For probabilistic

classification the default loss is log loss or Brier loss, and for regression it is MAE or RMSE. Accuracy, balanced accuracy, AUROC, subgroup error, and calibration are secondary views, and none of them substitutes for a proper primary loss.

If tuning is performed, outer assessment surrounds the complete inner selection. Define selection optimism on the loss scale as

$$O_b = \hat{\mathcal{R}}_{b,\text{outer}} - \hat{\mathcal{R}}_{b,\text{inner selected}}. \quad (3)$$

Positive O_b means the selected configuration looked better during search than it performed outside that search. A locked holdout supports uncertainty conditional on one fitted workflow. Nested cross-validation supports algorithm-level assessment when a single holdout would be too costly. These targets should be named rather than blended (Bates et al., 2024).

3.3 Robustness

Let $Q_{\lambda,r}$ be perturbation draw r at severity λ , calibrated on $\mathcal{T}_b$. It maps an audit case to $(X^{\lambda,r}, Y^{\lambda,r})$. The perturbation risk is

$$\mathcal{R}_{P,Q_{\lambda}}(f) = \mathbb{E}_P \mathbb{E}_{Q_{\lambda}} [\ell(f(X^{\lambda}), Y^{\lambda})], \quad (4)$$

and the excess response is $\Delta(\lambda) = \mathcal{R}_{P,Q_{\lambda}}(f) - \mathcal{R}_{P,Q_0}(f)$. For a constant-predictor reference L_{ref} , retained skill is

$$S(\lambda) = \frac{L_{\text{ref}}(\lambda) - \mathcal{R}_{P,Q_{\lambda}}(f)}{L_{\text{ref}}(0) - \mathcal{R}_{P,Q_0}(f)}. \quad (5)$$

StressFold reports the full $S(\lambda)$ curve, a prespecified degradation area, and the first severity at which $S(\lambda) \leq 1/2$, when crossed. Areas are never pooled across operators whose severity scales have different meanings.

Two label semantics are allowed. A *measurement perturbation* changes the representation while holding the latent case and label fixed. A *semantic perturbation* changes the case itself and requires a label oracle or known simulator. If neither claim is defensible, the protocol reports prediction movement or class-flip probability, not accuracy against a retained label.

3.4 Falsification

For exchangeability blocks C , the permutation null is

$$H_0 : Y \perp X \mid C.$$

Let $T(A, \mathcal{D})$ be a statistic for which higher values are better, produced by the complete pipeline. With M valid label permutations,

$$p = \frac{1 + \sum_{m=1}^M \mathbb{I}\{T(A, \mathcal{D}_m^{\pi}) \geq T(A, \mathcal{D})\}}{M + 1}. \quad (6)$$

Every permutation repeats preprocessing, selection, and tuning. The test asks whether the workflow found structure beyond the stated null (Ojala and Garriga, 2010). It

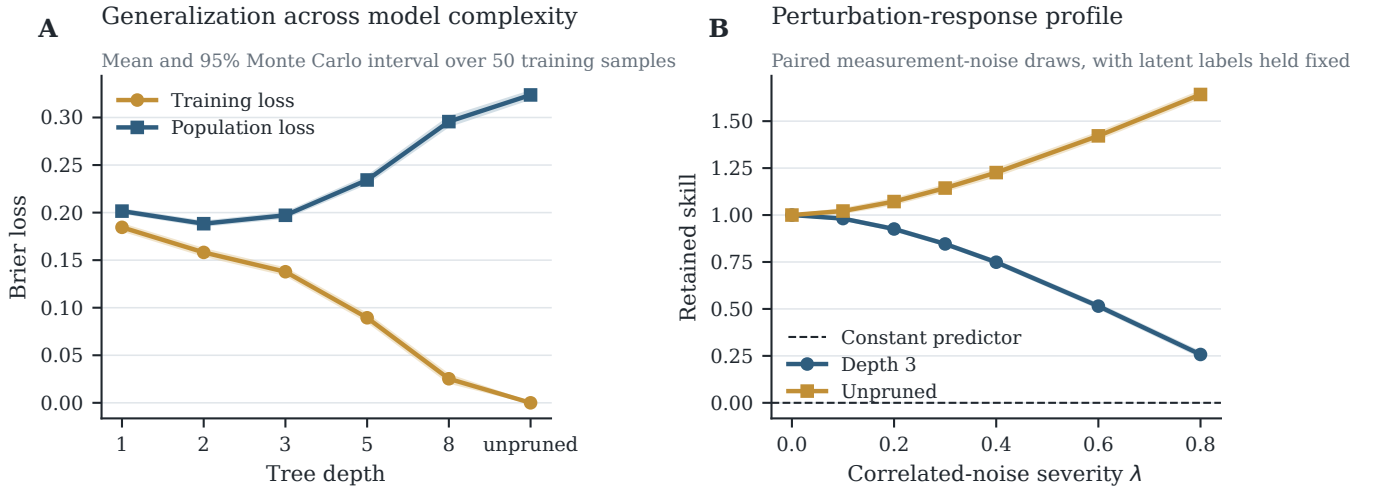


Figure 1: Generalization and robustness are distinct estimands. (A) Mean Brier losses and 95% Monte Carlo intervals across 50 independent training samples of size 300. Population loss uses 2,500 fresh cases per sample and the known Bernoulli probabilities. (B) Retained skill under correlated measurement noise for a fixed sample of 400 training and 3,000 latent audit cases, averaged over 40 common perturbation draws. Values above one indicate improvement relative to clean skill, which is a property of this particular perturbation and not a sign of superior generalization. The interpolating tree’s noise response illustrates why robustness cannot be used as a proxy for clean predictive risk.

does not show that the structure is causal, free of leakage, stable, or useful in a new domain. A separate random-label capacity curve corrupts a fraction of training labels, refits, and contrasts the training fit on corrupted labels with clean audit loss. That curve describes memorization capacity and carries no significance interpretation.

4 Perturbation operators

Every operator has a training-only calibration map, a severity scale with units, a constraint policy, and a declared label semantics. The defaults stay inside the observed support and avoid assumptions that cannot later be checked.

The covariate tilt avoids inventing impossible feature combinations. With

$$w_i(\delta, v) = \frac{\exp(\delta v^\top z_i)}{\sum_j \exp(\delta v^\top z_j)}, \quad n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2},$$

the direction v can be a domain feature, a subgroup contrast, or a principal component. A small n_{eff} warns that a nominally smooth shift is supported by few rows. Real site, time, or device partitions supersede this synthetic operator whenever available (Koh et al., 2021).

5 Paired protocol and aggregation

For each outer split, StressFold uses common audit cases and common random perturbations across candidate models. For model pair (a, c) , split b , severity λ , and draw r , define

$$D_{b\lambda r} = \hat{\mathcal{R}}_{b\lambda r}(f_a) - \hat{\mathcal{R}}_{b\lambda r}(f_c). \quad (7)$$

The split-level contrast is $\bar{D}_{b\lambda} = R^{-1} \sum_r D_{b\lambda r}$. Aggregation occurs over draws within split before summarizing over outer splits. Perturbed replicas of one audit row are not independent new observations, and the nominal sample size is never multiplied by R .

The practical protocol is:

1. Validate schema, constraints, row identity, grouping, and temporal order, then scan for exact and near duplicates across the intended split.
2. Construct outer splits. Inside each $\mathcal{T}_b$, fit all transforms and select hyperparameters. Freeze f_b .
3. Evaluate clean $\mathcal{H}_b$, then apply operators calibrated on the training fold to that same audit fold at a prespecified severity grid and seed schedule.
4. Refit A on cluster- or block-bootstrap versions of $\mathcal{T}_b$ and predict the unchanged $\mathcal{H}_b$ to measure refit instability.
5. Run label permutations within valid blocks, repeating the complete workflow. Run training-label corruption as a separate descriptive experiment.
6. Fit any replica generator on $\mathcal{T}_b$ only, and audit it against real $\mathcal{H}_b$ before using its rows as stress probes.
7. Export losses, paired contrasts, uncertainty, constraints, split indices, seeds, and software versions. Do not emit a single global score.

For a locked holdout, uncertainty in an additive loss difference is obtained by paired bootstrap resampling of audit cases after averaging their perturbation draws. For grouped or temporal observations, resampling respects

Table 1: Core operators. D contains training-fold robust scales and $LL^\top$ is a shrinkage correlation estimate. Operators are run separately so that each curve retains an interpretable horizontal axis.

Operator	Mechanism	Valid interpretation
Correlated noise	$x'_{\text{num}} = x_{\text{num}} + \lambda DL\epsilon$, $\epsilon \sim N(0, I)$. Clip only to declared physical bounds and report clipping.	Measurement robustness when the latent label is invariant to sensor error.
Categorical transition	Independently retain level a , or sample $a' \sim K_j(\cdot a)$, with transition mass controlled by λ .	Coding or observation error where transitions are plausible. Otherwise prediction stability only.
Missingness	Apply a mask at rate λ . MCAR is explicit. MAR requires a user-supplied mechanism.	Robustness to the named observation process, including the fitted imputer.
Tail contamination	Replace a prespecified fraction of cells or rows with empirical-tail values subject to schema constraints.	Sensitivity to outliers within observed support, not adversarial robustness.
Covariate tilt	$w_i(\delta, v) \propto \exp(\delta v^\top z_i)$. Resample complete observed rows and report effective sample size.	Reweighted empirical shift along a named risk direction, with observed labels still attached.
Training-label corruption	Replace a fraction q of training labels and rerun A . Never corrupt the clean audit labels.	Memorization and regularization response, not an estimate of deployment performance.
Local smoothed bootstrap	Resample a complete row, perturb numeric fields locally, and modify categories within a declared neighborhood.	Local support probing. Labels are retained only for features declared label-invariant.

clusters or blocks. Under repeated cross-validation, ordinary fold standard errors are not reported at all, because fold overlap induces dependence and dedicated nested-CV intervals are available (Bates et al., 2024). When many severities are used for formal testing, a prespecified maximum statistic or multiplicity correction replaces a row of unadjusted p -values. Response bands are descriptive by default.

6 Controlled checks of aggregation and falsification

Figure 2 isolates two protocol choices. First, paired perturbations give each model the same Monte Carlo difficulty. At $R = 16$, independent draws had 1.87 times the estimator standard deviation of common draws in the deterministic experiment. Pairing does not remove sample uncertainty, but it avoids paying unnecessary simulation variance in a model comparison.

Second, a permutation test must repeat selection. Under a pure null, 40 random binary predictors were scored on the same 200 cases and the best was selected. Permuting labels while holding that winner fixed produced a 77.3% false-positive rate at nominal 5%. Repeating the full 40-way selection inside each of 199 permutations reduced the empirical rate to 4.0% over 150 null repetitions. The example is deliberately stark, so that the conditioning error becomes visible without any appeal to asymptotics.

Claims built on permutation evidence should stay modest. A low p -value says that the workflow’s statistic is unusual under the encoded exchangeability null. A leaked target surrogate is also unusual under that null. Duplicate detection, group-aware splitting, time-aware split-

ting, and domain review are therefore controls in their own right, and treating them as optional preprocessing defeats the test.

7 Synthetic replicas are audited instruments

A learned generator $G(\mathcal{T}_b)$ is a random function of the same finite training sample. Its rows can expose local instability, probe constraints, or exercise software. They do not provide new information about unobserved modes of P , and they cannot enlarge the effective sample size of a confidence interval. A diffusion generator such as TabDDPM can model heterogeneous tables (Kotelnikov et al., 2023), but model complexity does not alter this information boundary.

The reference protocol therefore evaluates a generator along separate axes:

- **Fidelity.** Held-out classifier two-sample testing (C2ST) (Lopez-Paz and Oquab, 2017) and maximum mean discrepancy (MMD) with permutation calibration (Gretton et al., 2012).
- **Coverage.** Marginal, pairwise, subgroup, and target-conditional discrepancies, with support diagnostics in the style of precision and recall (Alaa et al., 2022).
- **Authenticity.** Exact copies, and train-to-synthetic versus holdout-to-synthetic nearest-neighbor structure.
- **Utility.** A fixed reference model trained on synthetic data and evaluated on the untouched real holdout (TSTR).

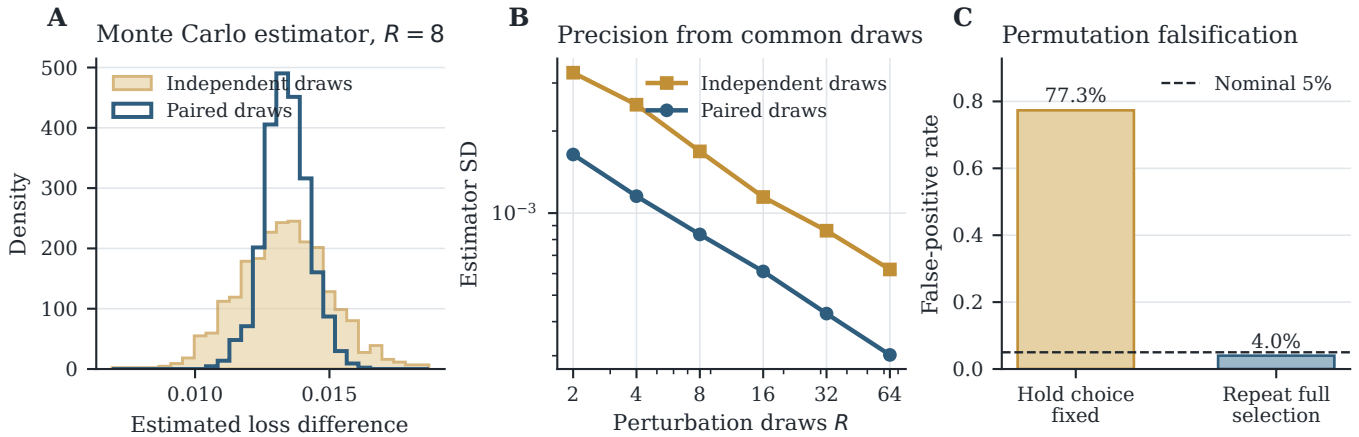


Figure 2: Pairing improves precision, and full-workflow permutation restores nominal calibration in this controlled simulation. (A) Sampling distributions of the depth-three minus depth-four loss contrast at severity $\lambda = 0.40$, using eight draws and a pool of 1,000 deterministic perturbations. (B) Estimator standard deviation over 1,000 Monte Carlo resamples at each R . (C) Null false-positive rates for a selected random predictor when model choice is held fixed versus repeated inside every permutation.

A C2ST AUC near one indicates a detectable difference. A value near one half means the classifier failed to distinguish the two samples, which is weaker than evidence of equality. MMD carries the same power caveat. Synthetic-only evaluation is never treated as unbiased evaluation against the real population, and methods that combine generated with real labels need an explicit correction design (Boyeau et al., 2025).

Figure 3 constructs the failure analytically. Draw $Y \sim \text{Bernoulli}(1/2)$, $S \in \{-1, 1\}$ uniformly, and

$$X_1 = S + \epsilon_1, \quad X_2 = (2Y - 1)S + \epsilon_2, \quad \epsilon_j \sim N(0, 0.3^2).$$

Each class has identical one-dimensional marginals but opposite dependence. Training contains 800 rows and the real holdout 3,000. The empirical bootstrap copies all generated rows, the smoothed bootstrap adds $N(0, 0.12^2)$ noise, and the marginal generator samples X_1 and X_2 independently within class. Table 2 shows why generator claims require more than one diagnostic. The C2ST value is reported without orientation, as $\max(\text{AUC}, 1 - \text{AUC})$.

The empirical bootstrap has excellent utility and dependence fidelity but every generated row is an exact copy. The smoothed bootstrap removes exact copies in this continuous example while retaining utility. Neither result is a privacy guarantee. The marginal generator appears acceptable under univariate checks yet fails both dependence and downstream utility. A strong audit keeps those conclusions separate rather than averaging them into a generator score.

8 Reproducibility and implementation boundary

The reference experiment script, `scripts/reproduce_paper.py`, is independent of

the package API. It uses NumPy, SciPy, CSV output that pandas can read, Matplotlib, and scikit-learn with a fixed root seed of 20260729 and single-threaded estimators. It regenerates every plotted PDF/PNG and both LaTeX/CSV tables. The DGP equations, sample sizes, replicate counts, model settings, and Monte Carlo schedules are encoded directly in that script.

The current interfaces implement a bounded subset of this reference protocol, and they do not embody every equation above. Browser protocol 0.3 uses independent standardized numeric noise, a protected retained-skill denominator, and a tie-adjusted descriptive null rank. Python package 0.2 reports a pooled paired null-exceedance rate across repeated holdouts and stops short of calling it a permutation p -value. Equation (6) requires coherent dataset-level permutations, a complete workflow rerun, and one prespecified aggregate statistic per permutation.

A production run should serialize more than an image. The record needs data and schema fingerprints, target and feature roles, constraints, exact split indices, estimator and preprocessing specifications, the seed tree, perturbation parameters, counts of rejected rows, software versions, and the raw per-case and per-draw losses. Report generation happens downstream of that record, which is never edited in place. A local-first browser interface can compute a bounded baseline directly, as the current lab does, or hand off to a richer local Python audit. Deep generators stay optional, and the browser never depends on them.

The controlled results are deterministic conditional on the documented software stack, subject to small floating-point effects and differences in library versions. The simulation intervals quantify repetition in the stated DGP, not epistemic uncertainty about whether that DGP resembles a user’s deployment environment.

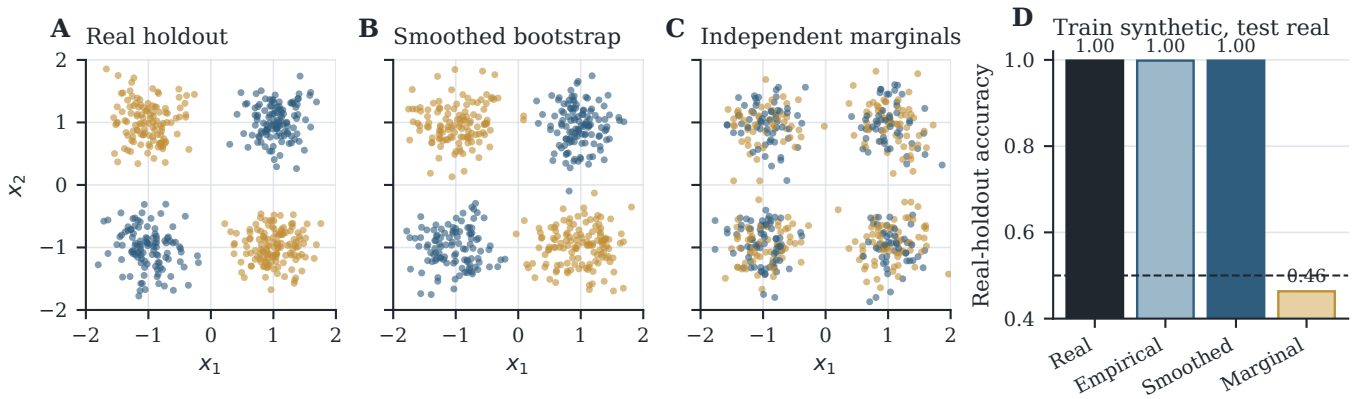


Figure 3: Marginal fidelity does not imply predictive fidelity. In the known DGP, class labels select opposite diagonal modes. A smoothed row bootstrap preserves that joint structure, while independently sampling each feature within class preserves conditional one-dimensional marginals but destroys dependence. The marginal generator’s mean conditional KS discrepancy is only 0.069, yet its real-holdout TSTR accuracy is 0.463. Bar labels are rounded, and the dashed line marks chance.

Table 2: Synthetic-replica diagnostics. KS is the mean class-conditional feature statistic. Correlation error is the mean absolute class-conditional correlation difference. C2ST compares the joint (X, Y) distribution. TSTR is real-holdout accuracy.

Replica	KS ↓	Corr. err. ↓	C2ST AUC ↓	TSTR ↑	Copies ↓
Real training data	0.000	0.000	0.500	0.998	0.0%
Empirical bootstrap	0.058	0.013	0.598	0.998	100.0%
Smoothed bootstrap	0.056	0.024	0.553	0.998	0.0%
Independent marginals	0.069	0.910	0.777	0.463	0.1%

9 Limitations and open questions

Operator validity is the main limitation. Gaussian noise, empirical tilts, and local resampling are convenient, not universal. A response curve is only as relevant as its mechanism. Natural held-out domains should be used whenever available, and failure to observe a natural shift must not be translated into evidence of robustness.

Labels may cease to be valid. Large feature perturbations can move a row across the true decision boundary. Without a simulator or label oracle, accuracy against the retained label then mixes model error with annotation invalidity. StressFold falls back to prediction consistency and reports the absence of a correctness claim.

Exchangeability can fail. Permutation tests, ordinary bootstrap, and random splitting all rely on a sampling unit. Households, patients, devices, sites, and time intervals require explicit blocks. A valid blocked null may have few permutations and low power.

Generator diagnostics are incomplete. C2ST and MMD can miss differences in high dimensions or small samples. A nearest-neighbor copy check is not a privacy proof, and TSTR depends on which reference model is chosen. The fidelity, coverage, authenticity, and utility dimensions proposed for generative-model auditing (Alaa et al., 2022) complement each other without being exhaustive.

The protocol detects risk signals, not causes. A large train-audit gap can arise from variance, shift, leakage, or an inappropriate split. A steep perturbation curve can reflect a genuinely hard measurement problem. The evidence profile narrows the hypotheses, and domain investigation determines the remedy.

The extensions worth doing next are empirical ones: block-aware operators for longitudinal data, user-defined invariance contracts, simultaneous confidence bands with explicit multiplicity targets, and benchmark suites with predeclared DGPs, leakage traps, and expected operating characteristics.

10 Conclusion

The question “is this model overfit?” has no single auditable answer. StressFold replaces it with several questions that do, each carrying its own estimand. Clean outer assessment measures predictive generalization under a split policy. Perturbation-response curves measure sensitivity to named changes. In the reference protocol, full-workflow permutation and random-label experiments test prespecified null and memorization hypotheses, while audited synthetic replicas supply controlled probes.

At release 0.2, the browser and Python interfaces implement the core clean and stress workflows with descrip-

tive null controls. Full-workflow permutation inference and generator auditing remain reference-protocol extensions.

The controlled experiments show why the separation matters. Clean population loss and noise response can move in opposite directions. Paired Monte Carlo makes comparisons more precise. An incomplete permutation invalidates the inference it appears to support. Good marginals can coexist with synthetic dependence that is useless downstream. A report that keeps these four results apart is harder to summarize than a single score, and it is the only form in which the evidence can be checked.

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